UNIT 2 · RAPID REVISION

Formula and Fact Sheet

Machinery elements (Topics 1–2)

Item	Value
Flat rubber belt k / tension ratio (180°)	0.25–0.30 / 2.2–2.6
V-belt design tension ratio	5
V-belt section included angle / groove angle	40° / 34–38°
V-flat balanced arcs	130° (sheave) vs 230° (flat pulley)
Max belt power condition	$T_1 = 3T_c$; v_opt = $\sqrt{(T_1/3m)}$
Agricultural V-belt sections	HA, HB, HC, HD, HE (+HAA, HBB double)
Min sprocket teeth (high speed)	17–19 (chordal action)
Chain speed	$v = p \cdot T \cdot N/60$
Min gear teeth, φ=20°	17 $(2/\sin^2\varphi)$
Module	$m = D/T$; $p_c = \pi m$
Lewis equation	$F_t = \sigma_b \cdot b \cdot Y \cdot m$
Bearing life	$L_{10} = (C/P)^p \times 10^6$ rev; p = 3 ball, 10/3 roller
L_{10} meaning	90% survival life
Wahl factor	$(4C-1)/(4C-4) + 0.615/C$
Spring index C	D/d, ideally 6–12
Bolt class 8.8	UTS 800 MPa, yield 640 MPa
PTO splines	6 (540 rpm), 21 (1000 rpm)
Single U-joint speed ratio range	cos α to 1/cos α per rev
Overrunning clutch	one-way freewheel — high-inertia PTO rotors
Slip clutch vs shear pin	auto-reset friction vs one-time breakage

System characteristics, balancing and stability (Topics 3–4)

Item	Key point
System capacity	Limited by weakest functional component
Reliability of series system	Product of component reliabilities
Draft–speed relation (MB bodies)	$D = a + bV^2$
Static balance	$\Sigma F = 0$ (CG on axis)
Dynamic balance	$\Sigma F = 0$ and $\Sigma M = 0$; two correction planes
Centrifugal force	$F = m\,r\,\omega^2$ ($\propto$ speed^2)
Reciprocating force	Cannot be balanced by rotating mass
Mower compromise	Counterweight = $F_c + \frac{1}{2} F_{r,max}$ ("half balancing")
Full balance options	Opposed reciprocating counterweight / two opposed knives
Double-knife benefit	Half stroke → higher crank rpm → higher forward speed
Flywheel	$\Delta E = I \cdot C_s \cdot \omega^2$; smooths cyclic loads (baler plunger)
Resonance	Excitation ≈ natural frequency $\sqrt{(k/m)}$
Lateral overturn angle	$\tan\theta = B/2h$
Front axle minimum load	≈ 20% of tractor weight (mounted implement)

Sowing, fertilizer and inter-cultivation design (Topics 6–7)

Item	Key point
Precision metering targets	95–100% cell fill; low plate speed
Best plate speed (Barmington data)	~50 fpm (0.25 m/s); range 35–100 fpm
High forward speed with plates	More cells/larger plate, NOT faster plate
Metering unit musts	Cutoff device + positive unloading
Fluted roller rate control	Exposed flute length (near-linear)
Helical flutes	Steadier, more uniform flow than straight
Planter spacing	$s = \pi D_w/[n_c \cdot i \cdot (1-\sigma)]$ — σ = drive-wheel skid; skid widens spacing
Hopper wall slope	> angle of repose (design ≥35–40° for grain)
Double-disc opener angle	7–10° between discs
Fertilizer metering challenge	Hygroscopic + corrosive + variable flow
Star-wheel feed	Grain-drill fertilizer units; gate + wheel size sets rate
Spinner pattern	Triangular; 25–35% overlap for uniformity
Band placement	5 cm beside × 5 cm below seed
Cultivation chief benefit	Weed control (Kepner)
Sweep included angle	60–90°, low pitch, 3–5 cm deep
Spring-trip force	~1.5–2× working draft, auto-reset
Rotary hoe	Ground-driven spiders, 8–16 km/h, blind cultivation
Drill calibration	$N = A/(\pi D W n)$; CV ≤ 5–7%

Force analysis, pull–draft–power and furrow openers (Topics 5–6)

- Draft = $P \cdot \cos\theta$ (horizontal, in travel direction); vertical component = $P \cdot \sin\theta$ helps tractor traction, hurts implement penetration.
- Unit draft = $D/(w \cdot d)$ — a pressure; 1 kgf/cm² ≈ 98.1 kPa ≈ 0.0981 N/mm².
- $P(kW) = D(kgf) \cdot v(km/h)/367 = D(kN) \cdot v(m/s)$; $P(hp) = D(kgf) \cdot v(m/s)/75$.
- Goryachkin: $P = fG + kab + \varepsilon abv^2$ — only the LAST term depends on speed (as v^2).
- ASAE D497: $D = Fi(A + B \cdot S + C \cdot S^2) \cdot w \cdot d$; MB plough A=652, B=0, C=5.1; F(fine)=1.0, F(medium)=0.70, F(coarse)=0.45.
- Centre of resistance (MB bottom): ~¼ width from landside, ~½ depth below surface; pull line through it → no side draft.
- MB plough penetrates by suction; disc plough by weight (soil reaction on disc has upward component).
- Zero-till residue sowing → inverted-T opener; precise placement on planters → double-disc; hard dry soil → hoe type; clean moist seedbed → runner; trashy → shoe/disc.

Harvester and thresher design (Topic 8)

- Cutter bar: pitch = guard spacing = stroke = 76.2 mm; crank 800–1000 rpm = 1600–2000 cuts/min; v_max = ωr at mid-stroke; register at stroke ends; lead ~1–2 cm/m of bar.
- Impact cutting (rotary mowers) needs minimum knife speed; shear cutting is knife + ledger plate.
- v = πDN/60; wheat rasp-bar 25–30 m/s; wire-loop paddy ~12–15 m/s; rubber rolls 1.3–1.5 m/s; damage ∝ ~v², unthreshed loss falls with v.
- Concave clearance tapers: wide at feed end → narrow at rear. Threshing efficiency = 100 - % unthreshed.
- Combine losses: header (often largest, shatter), cylinder (unthreshed), walker (free seed in straw), shoe (blown over). Tailings return to cylinder.
- Reel index 1.25–1.5; pickup reel for lodged crop; axial-flow = multiple gentle impacts, standard for paddy.
- IS 9129 — thresher feeding-chute safety standard (withdrawn by BIS; now covered by IS 9020).

Controls, mechatronics, ML, IoT, robotics and drones (Topics 9, 10, 12)

- F = p·A; P(kW) = p(bar)·Q(L/min)/600; tractor pump = external gear, open-centre circuit; relief valve = max-pressure protection.
- Vacuum plate planter = negative pressure singulation; air seeder = central hopper + air-stream distribution.
- ISOBUS = ISO 11783 (CAN-based tractor–implement interface); mechatronic loop = sensor → ECU → actuator → feedback.
- Supervised = labelled data; CNN = images (weed detection, See & Spray); LoRaWAN = long-range low-power IoT; NDVI = (NIR-R)/(NIR+R); RTK-GNSS = 2–3 cm.
- Drone classes: nano <250 g, micro ≤2 kg, SMALL 2–25 kg (agri), medium ≤150 kg, large >150 kg; green zone ≤120 m; Digital Sky = DGCA platform; UIN + RPC required.
- Drone spray (crop-specific SOP, 2023): ~20–25 L/ha by crop stage; ~1.5–2.5 m above canopy (3–4 m sugarcane); ~3.5–5 m/s; no spraying in wind >~10 km/h; LiPo batteries.
- Kisan drone: ICAR/KVK/SAU 100% (≤₹10 lakh), FPO demo 75%, CHC 40%/graduate 50%, individual 40–50%; Namo Drone Didi: 80% ≤₹8 lakh to 15,000 women SHGs.

Further reading

- Kepner, Bainer & Barger — Principles of Farm Machinery: Ch. 1–2, 4, 11–14, 17.
- Machine design: Shigley; V.B. Bhandari. Jagdishwar Sahay — Elements of Agricultural Engineering.
- ASAE Standard D497 (Agricultural Machinery Management Data); ISO 11783 (ISOBUS).
- Drone Rules 2021 (Ministry of Civil Aviation); MoA&FW SOPs for drone application of pesticides & nutrients; SMAM guidelines (agriwelfare.gov.in).
- NPTEL lectures: Farm Machinery & Equipment; mechatronics & precision agriculture.